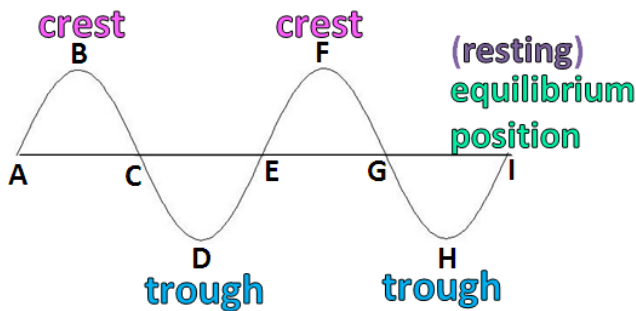
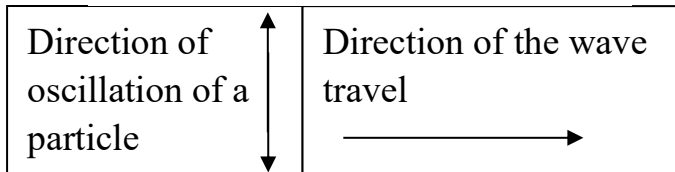
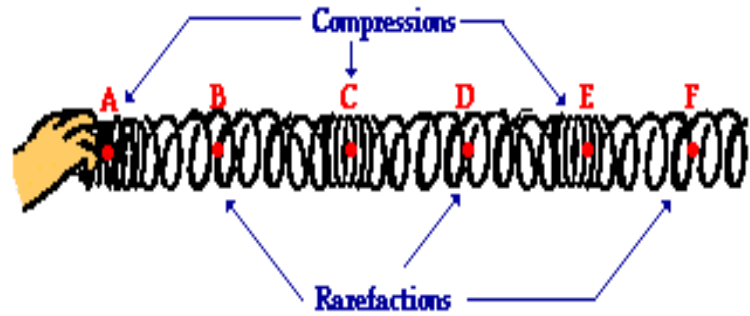
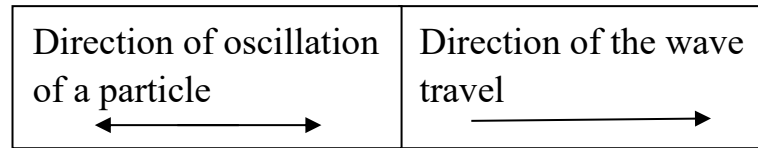


Wave motion

Transverse wave



Longitudinal wave



Transverse waves

- Direction of **oscillation** of a particle in the medium is **perpendicular** to the direction of **propagation** (wave travel).
- Crests and troughs are formed in the wave.

Longitudinal Wave

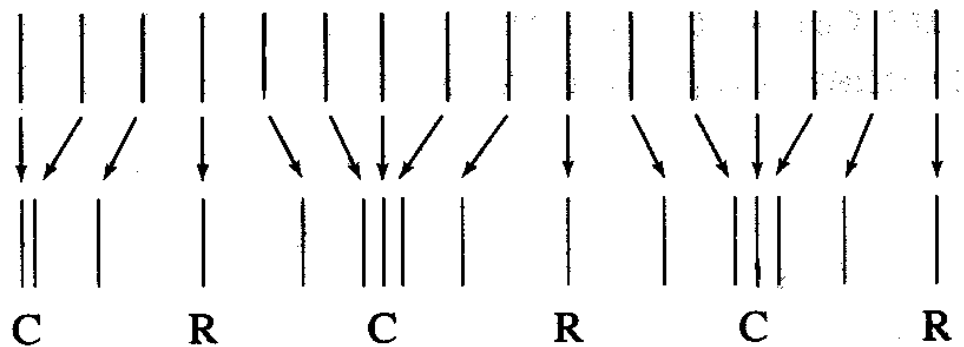
- Direction of **oscillation** of a particle in the medium is **parallel** to the direction of **propagation** (wave travel).
- Compressions and rarefactions are formed in the wave.

C : Compression
R : Rarefaction

Center of compression move at the **same direction** as the wave travel in the next second

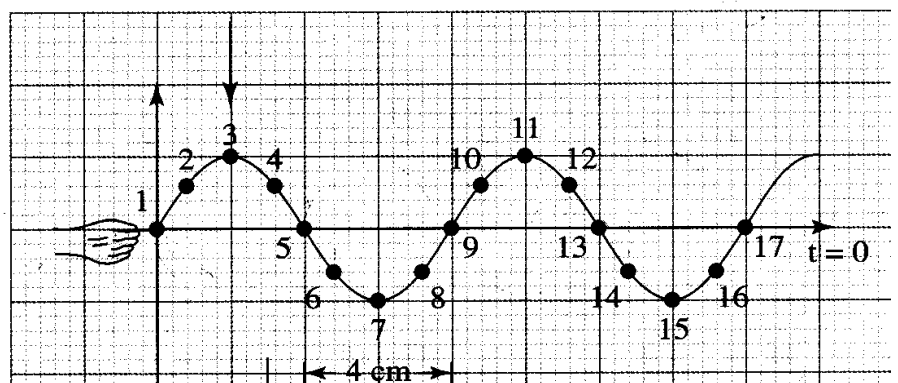
Equilibrium position

Positions of particles in a longitudinal wave

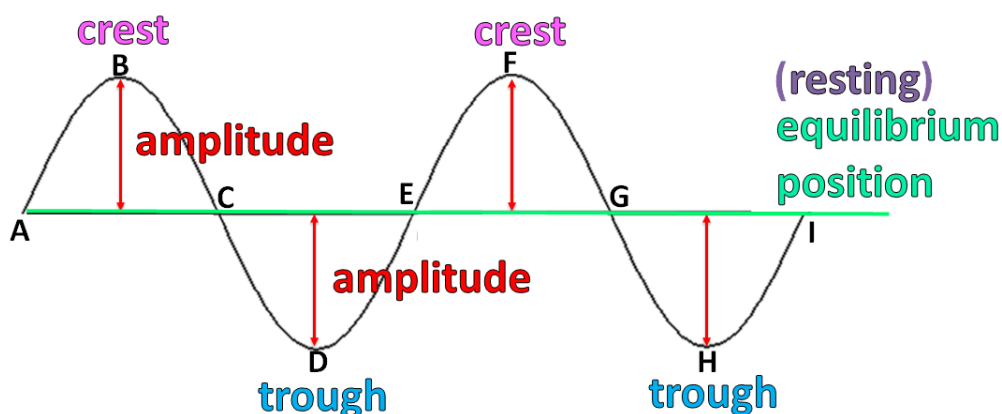


1. What kind of wave is generated in the string?

Answer :
Transverse wave

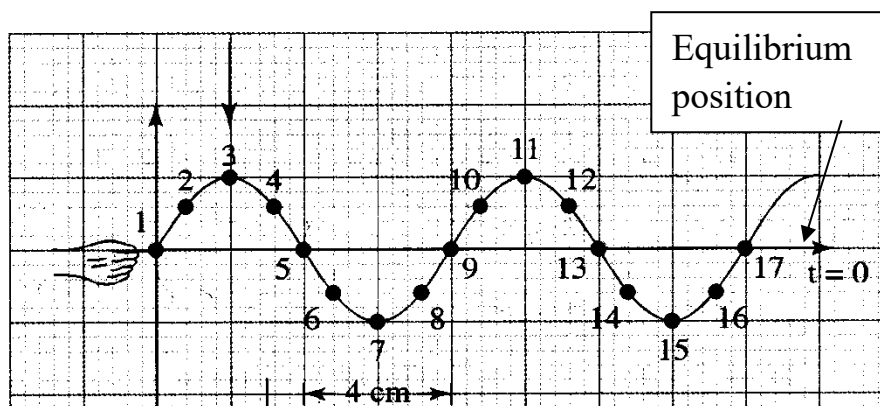


Amplitude (A) is the maximum displacement of any particle from its equilibrium position.

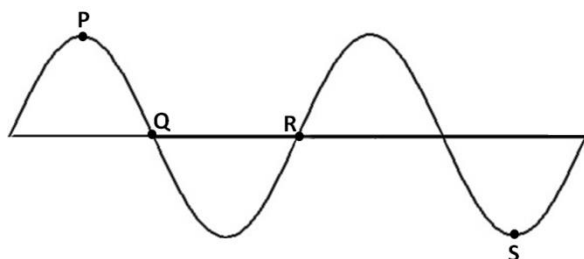


2. What is the amplitude of particle 3?

Answer : 2 cm



3. The following figure shows a wave travelling towards the right along a string. Consider four particles P, Q, R and S on the string.

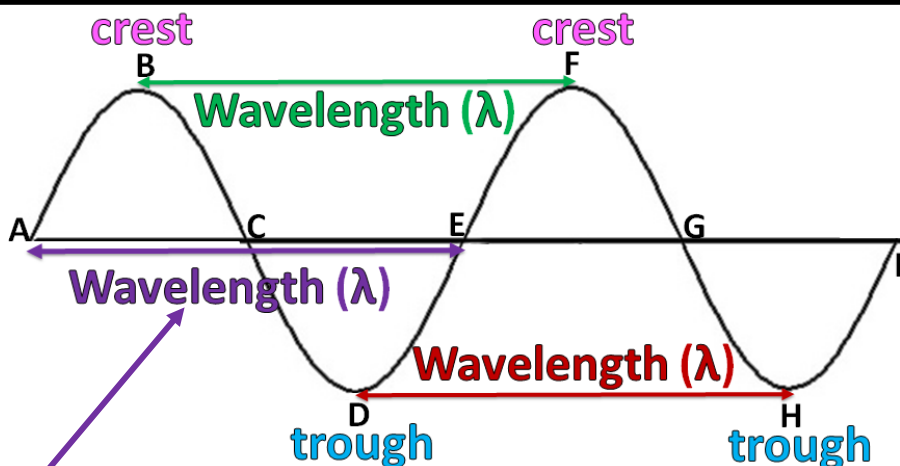


D

Which particle(s) has/have the same amplitude as P?

- A Q
- B R
- C S
- D Q, R and S

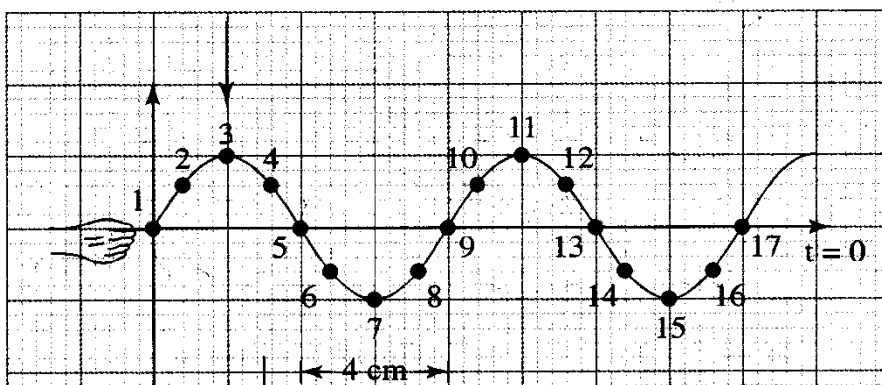
Wavelength (λ) is the distance between two successive **crests** or two successive **troughs**.
(Unit: m)



Wavelength (λ) is the distance between a particle and the nearest one which has the **same displacement and direction of motion** at any instant.

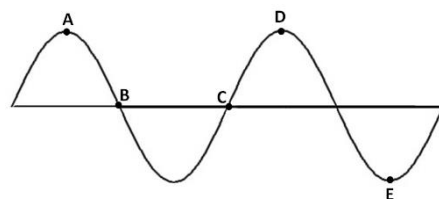
4. What is the wavelength of the wave?

Answer : 8 cm



5. The following figure shows a wave.
Which of the following is/are the wavelength of the wave?

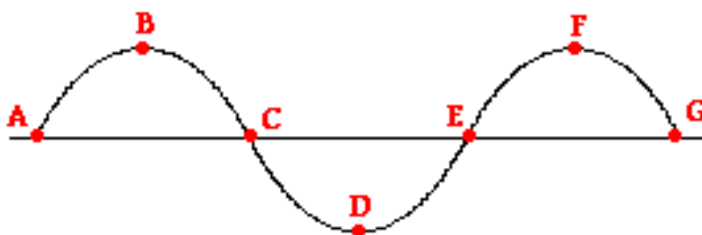
- A. AC
B. DE
C. CD
D. AD



D

6. Indicate the interval that represents one wavelength.

- A. A to C
B. B to D
C. A to G
D. C to G



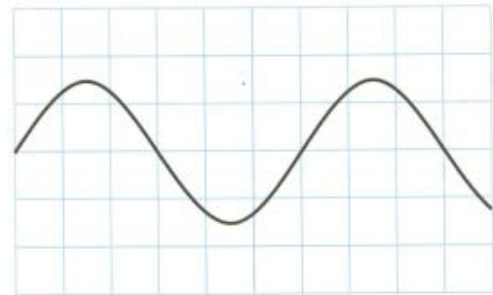
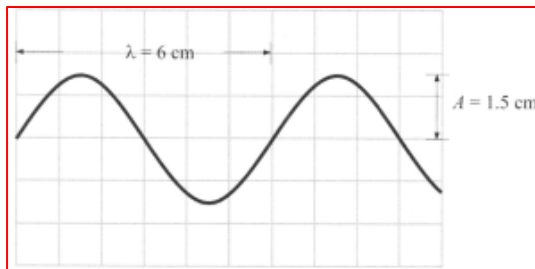
D

(For question 7 to 8)

The diagram shows a wave profile at a certain instant. The side of each square is 1 cm in length

7. The wavelength of the wave is

- A. 2 cm
B. 3 cm
C. 4 cm
D. 6 cm



D

8. The amplitude of the wave is

- A. 1 cm
B. 1.5 cm
C. 2 cm
D. 3 cm

Answer is B.

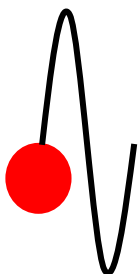
The amplitude is $A = 1.5$ cm.

B

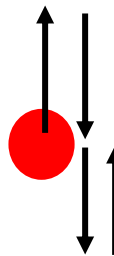
Period (T) is the time taken for a particle to make one complete oscillation. It is also the time for the wave to advance one wavelength.

Unit: second (s)

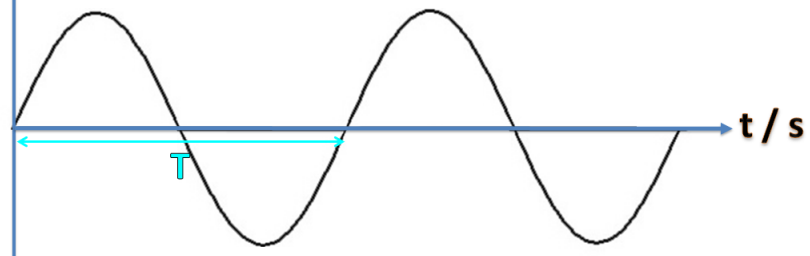
Complete oscillation



=



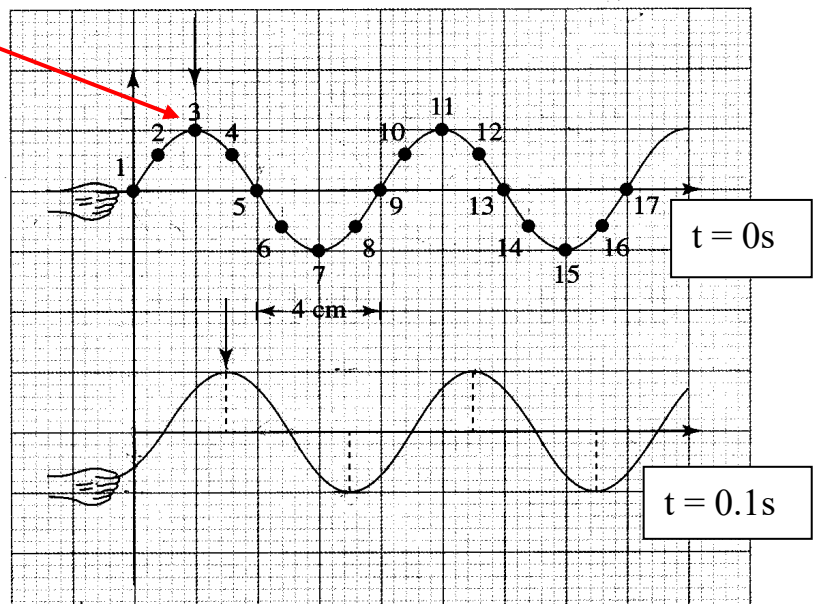
s / m



9. Where will be the crest (the one with arrow) after $t = 0.8$ s?

What is the period (T) of the wave?

Answer : 0.8 s



Frequency (f) is the number of oscillations made by any particle in one second.
Unit: Hertz (Hz)

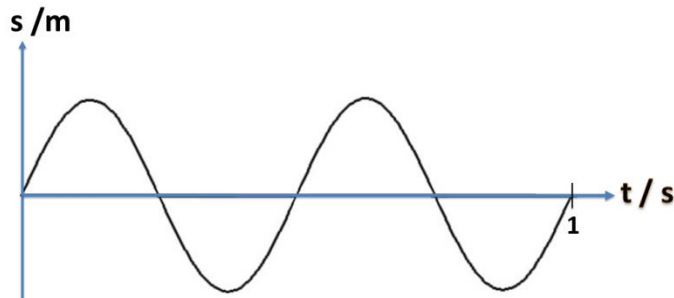
$$f = \frac{1}{T}$$

Wave speed is given by $v = f\lambda$. Unit: ms^{-1}

10. Find the period and the frequency of the wave.

$$T = 0.5 \text{ s}$$

$$F = 1/T = 2 \text{ Hz}$$



11. If the wavelength and the frequency of a wave are 0.2 m and 5 Hz respectively, what is the speed of the wave?

A 0.1 m s^{-1}

B 0.5 m s^{-1}

C 1 m s^{-1}

D 5 m s^{-1}

C

12. The frequency of a wave is 3 Hz. How many complete waves can be produced in 2 s?

A 3

B 4

C 5

D 6

$$T = \frac{1}{3} \text{ s} \quad \text{Number of wave produced} = \frac{2}{\frac{1}{3}} = 6$$

D

13. A transverse wave travels along a string with a speed of 1.2 ms^{-1} . The diagram shows the shape of the string at a certain instant.

(1) Its wavelength is 20 cm

(2) Its frequency is 6 Hz

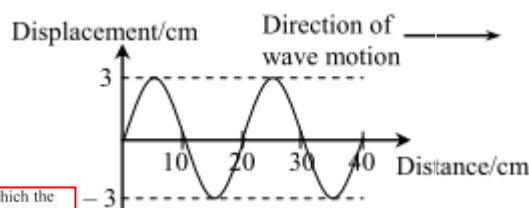
(3) Its amplitude is 6 cm

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)



A

- (1) ✓ The wavelength of a wave is the minimum distance in which the wave repeats itself.
- (2) ✓ By $v = f\lambda$
 $1.2 = f(0.2)$
 $\therefore f = 6 \text{ Hz}$
Note: The unit of λ must be changed to meter (m).
- (3) ✗ The amplitude of a wave is the maximum displacement of a particle (on the wave) from its equilibrium position. Hence, the amplitude is 3 cm.

14. A transverse wave is travelling along a string. The figure shows the variation of the displacement of a particle on the string. Which of the following is/are correct?

(1) The wave is travelling towards the left

(2) The frequency of the wave is 1.25 Hz

(3) The amplitude of the wave is 4 cm

A. (1) only

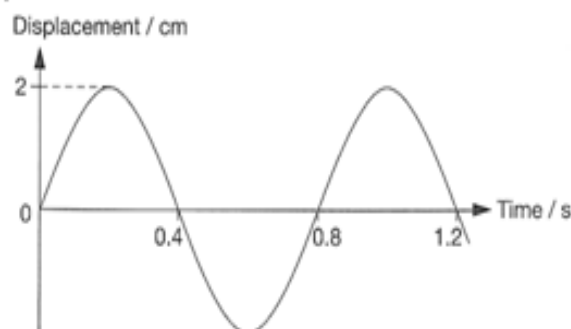
B. (2) only

C. (1) and (3) only

D. (2) and (3) only

B

- (1) Cannot be determined
(2) $f = 1/0.8 = 1.25 \text{ Hz}$
(3) $A = 2 \text{ cm}$

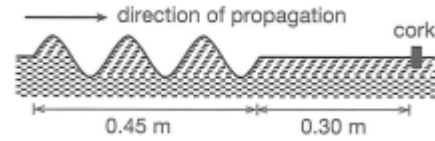


B

15. A series of water waves is travelling with a speed of 0.5 ms^{-1} towards a cork at the water surface as shown. At what time will the cork reach the highest position for the second time?

- A after 0.9 s
B after 0.975 s
C after 1.05 s
D after 1.125 s

Note that the cork will reach the highest position for the second time when water wave of 1.75λ passes through it.
Total distance
 $= 0.30 + 1.75 \times \frac{0.45}{3}$
 $= 0.5625 \text{ m}$
 $\therefore$ The time is at $t = \frac{0.5625}{0.5} = 1.125 \text{ s}$.

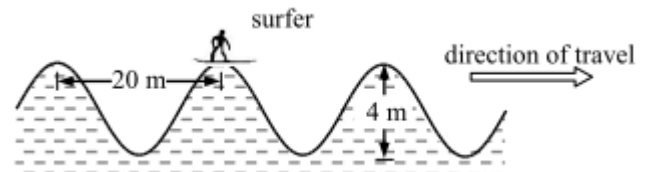


D

16. 2016 DSE MC Q.16

The surfer in the figure reaches a crest at the moment shown. The crests of the water wave are 20 m apart and the surfer descends a vertical distance of 4 m from a crest to a trough in a time interval of 2 s. What is the speed of the wave?

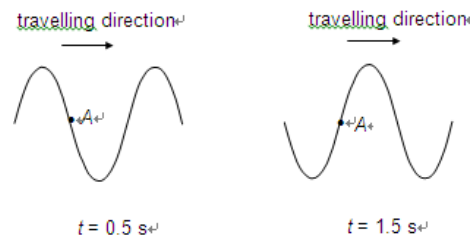
- A. 1 m s^{-1}
B. 2 m s^{-1}
C. 5 m s^{-1}
D. 10 m s^{-1}



C

17. A wave is travelling to the right. The waveforms at $t = 0.5 \text{ s}$ and $t = 1.5 \text{ s}$ are shown in the following figure. What is the minimum possible frequency of the wave?

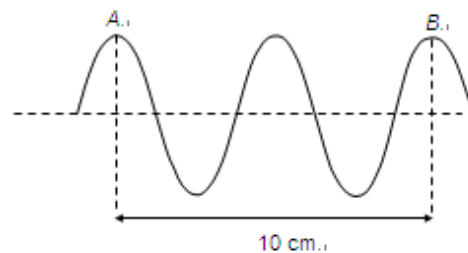
- A 0.25 Hz
B 0.5 Hz
C 0.75 Hz
D 1 Hz



B

18. If it takes 3 s for the peak at A to travel to B, what are the speed and frequency of the wave?

	Speed / m s^{-1}	Frequency / Hz
A	0.0167	0.333
B	0.0167	0.0167
C	0.0333	0.667
D	0.0333	0.0166

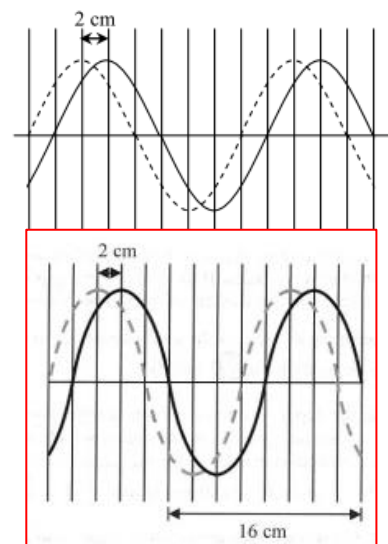


C

19. The solid curve in the diagram shows a transverse wave at a certain instant. After 0.05 s, the wave travelled a distance of 2 cm and is indicated to the dashed curve. Find the wavelength and frequency of the wave.

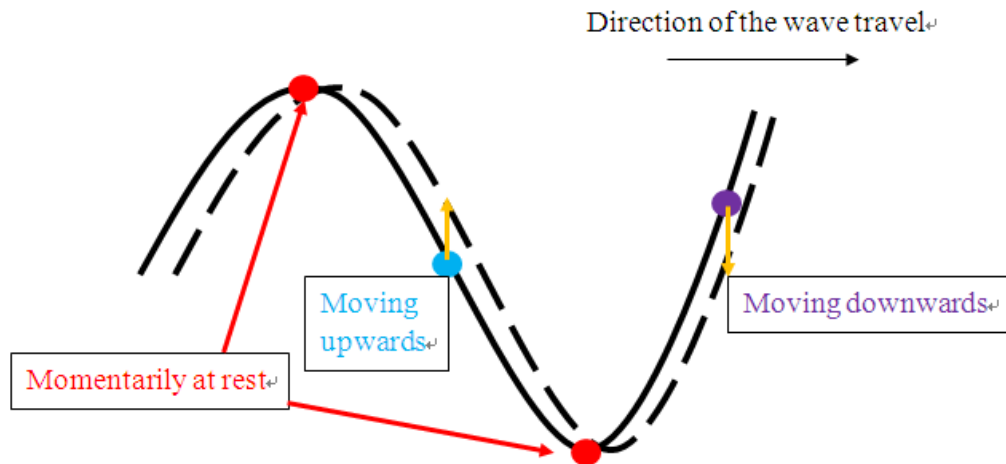
	Wavelength / cm	Frequency / Hz
A	8	2.5
B	8	5
C	16	2.5
D	16	5

The wavelength of a wave is the minimum distance that the wave repeats itself. It is equal to $8 \times 2 = 16 \text{ cm}$.
Since the wave travelled a distance of 2 cm in 0.05 s, the wave speed is equal to 0.4 m s^{-1} ($= \frac{0.02}{0.05} \text{ m s}^{-1}$).
By $v = f\lambda$, the frequency of the wave $= f = \frac{v}{\lambda} = \frac{0.4}{0.16} = 2.5 \text{ Hz}$
Note: The unit of λ must be changed to SI unit, i.e. meter (m).



C

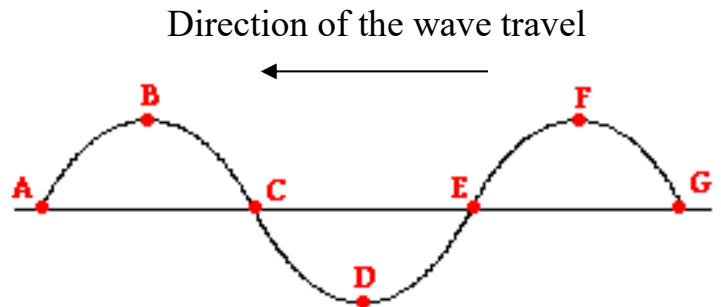
Direction of motion of particles



20. Identity all the particles that are

- (i) moving upwards?
- (ii) moving downwards?
- (iii) momentarily at rest?

- (i) A, E
- (ii) C, G
- (iii) B, D, F



Phase

In phase: Two particles separated by **whole number** of wavelength ($n\lambda$)

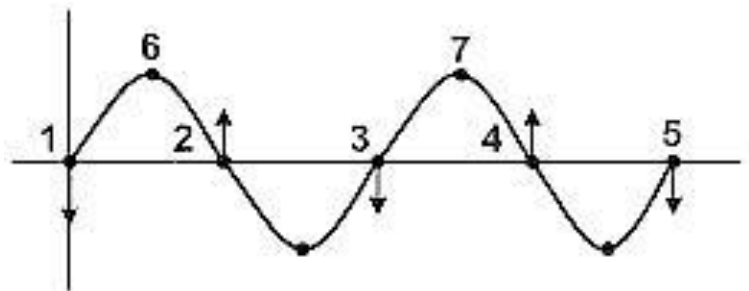
e.g. 1&3 (λ), 2&4 (λ), 1&5 (2λ)

Out of phase: Two particles that are **not** separate by whole number of wavelength ($n\lambda$)

e.g. 1&6 ($1/4\lambda$), 2&7 ($3/4\lambda$)

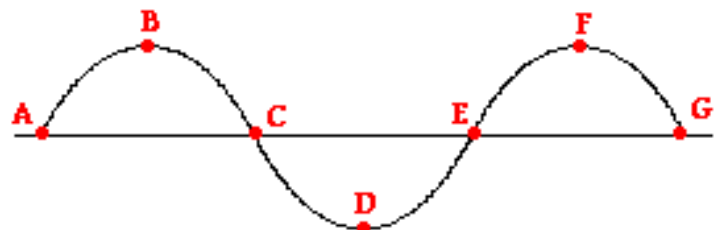
Antiphase: Two particles separated by $(n + \frac{1}{2})\lambda$

e.g. 1&2 ($1/2\lambda$), 2&5 ($1 + 1/2\lambda$)



21. Identity all the pairs of particle that are vibrating

- (i) in phase?
- (ii) out of phase?
- (iii) antiphase?



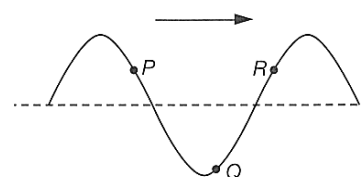
- (i) A&E, B&F, C&G
- (ii) A&B, A&D, A&F, B&C, B&E, B&G, C&D, C&F, D&E, D&G, E&F, F&G
- (iii) A&C, A&G, B&D, C&E, D&F, E&G

22. The figure shows a transverse wave propagating to the right. The time taken for the labelled particles to reach their respective equilibrium are different. Arrange the time taken in ascending order.

- A. P, Q, R
B. P, R, Q
C. Q, R, P
D. R, Q, P

D

Only R is moving towards the equilibrium position.
So it takes the shortest time. Q is almost at its peak.
So it takes shorter time to the equilibrium position than P.

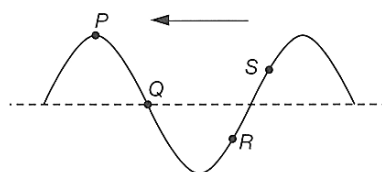


D

23. The figure shows the instantaneous position of the particles P, Q, R and S on a transverse wave propagating to the left. Which of the particles has its displacement and velocity in opposite directions at the instant shown

- A. P
B. Q
C. R
D. S

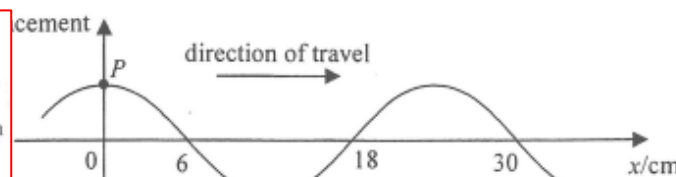
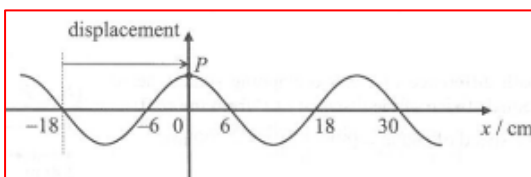
P is momentarily at rest and the displacement of Q is zero. So P and Q cannot be the particle.
The displacement of R is downwards but its velocity is upwards.
The displacement of S and its velocity are both upwards.



C

24. The figure shows a snapshot of a section of a continuous transverse wave travelling along the x-direction at time $t = 0$. At $t = 1.5$ s, the particle P just passes the equilibrium position for a second time at that moment. Find the wave speed.

- A. 20 cm s^{-1}
B. 12 cm s^{-1}
C. 6 cm s^{-1}
D. 4 cm s^{-1}



As shown in the figure above, the wave has travelled 18 cm when particle P just passes the equilibrium position for a second time.

$\therefore$ The wave speed $= 18 \div 1.5 = 12 \text{ cm s}^{-1}$

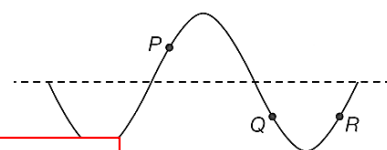
B

25. The figure shows a transverse wave travelling along a string. At the instant shown, particle P is moving downwards. Which of the following statements is incorrect?

- A. The wave is travelling towards the right
B. All the particles have the same amplitude.
C. Particle Q is moving upwards.
D. Particle R is moving in phase with P

D

The phase difference θ between P and R is in the range of $\pi < \theta < 2\pi$.



D

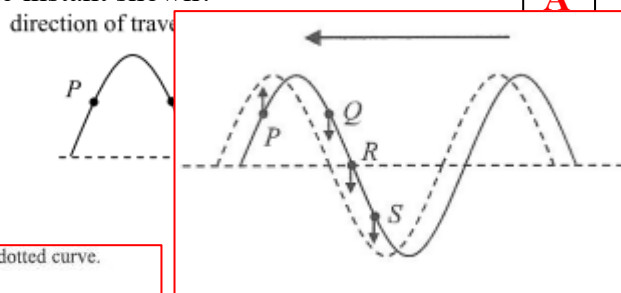
26. A transverse wave travels the left on a long string. P, Q, R and S are particles on the string. Which of the following statements correctly describes their motion at the instant shown?

- (1) P is moving upwards
(2) Q and S are moving in opposite directions
(3) R is momentarily at rest.

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

After a while, the wave would shift to the left shown as the dotted curve.

- ✓ (1) Particle P is moving upwards.
* (2) Particles Q and S should move in the same direction.
* (3) Particle R is not momentarily at rest, but moving downwards.

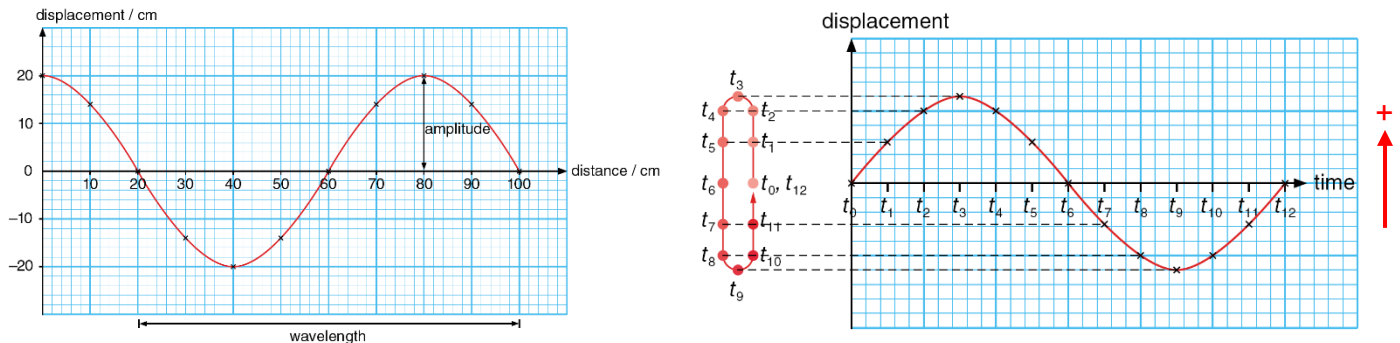


A

Graphical description of wave motion

The motion of vibrating particles in a traveling wave can be described by two types of graphs:

1. **Displacement – distance graph** (Shows displacement of **different particles** at **particular instant**)
2. **Displacement – time graph**. (Shows displacement of **particular particle** at **different instants**)



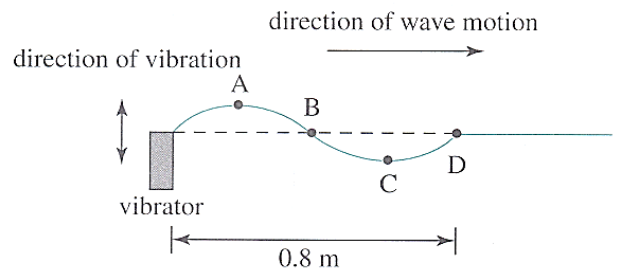
27. A vertical vibrator generates waves on a string. It takes 0.25 s to produce a complete wave of wavelength 0.8 m on the string.

- (a) Find the frequency and speed of the waves on the string.

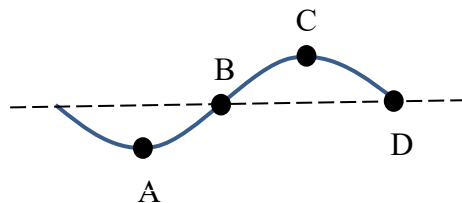
$$(a) T = 0.25 \text{ s}$$

$$f = \frac{1}{0.25} = 4 \text{ Hz}$$

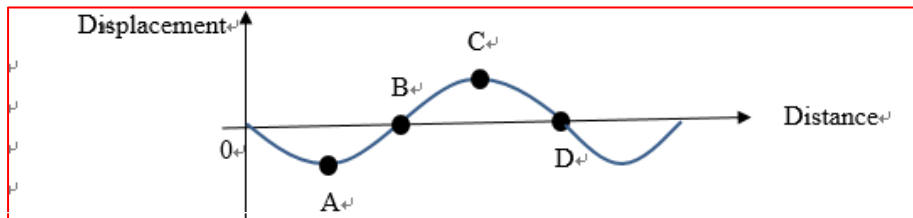
$$v = f\lambda = 4 \times 0.8 = 3.2 \text{ ms}^{-1}$$



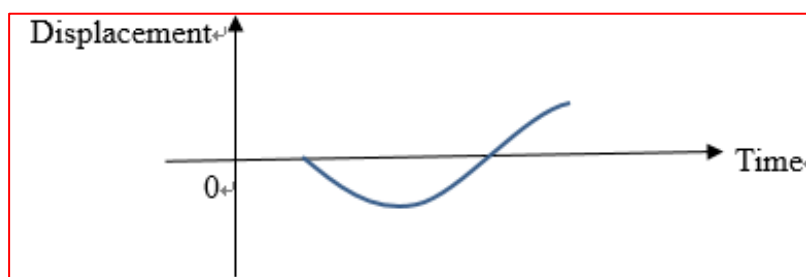
- (b) Sketch the shape of the string 0.125 s later, showing the position of particles A, B, C and D.



- (c) Sketch the displacement-distance graph of the string for $t = 0.375 \text{ s}$.

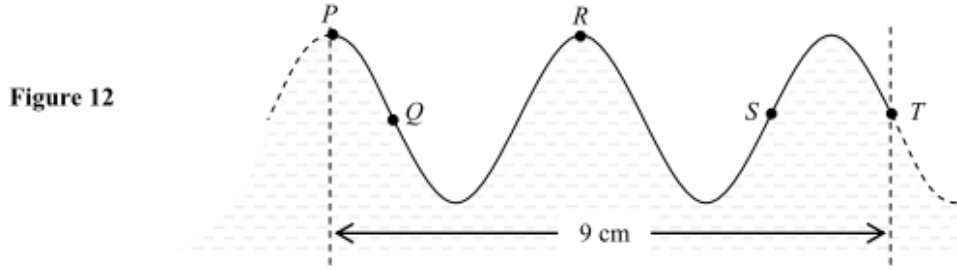


- (d) Sketch the displacement-time graph of the string for particle A.



28. 2009 CE Q.10

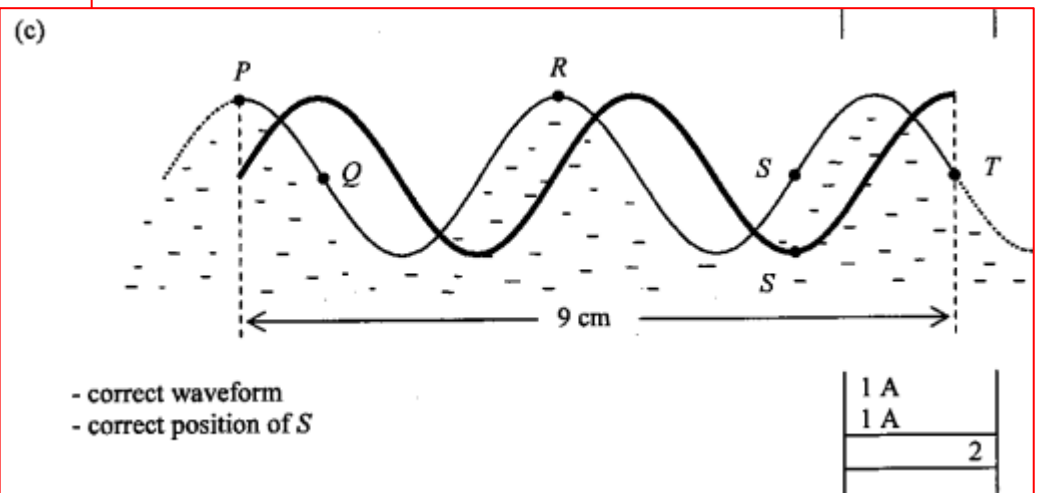
In a ripple tank, initially five tiny plastic beads (P, Q, R, S, T) float on the calm water surface. A vibrator begin to produce straight waves at time $t = 0$. Figure 12 shows the position of beads on the waves at $t = 7$ s. Figure 13 shows the displacement – time graph of S.



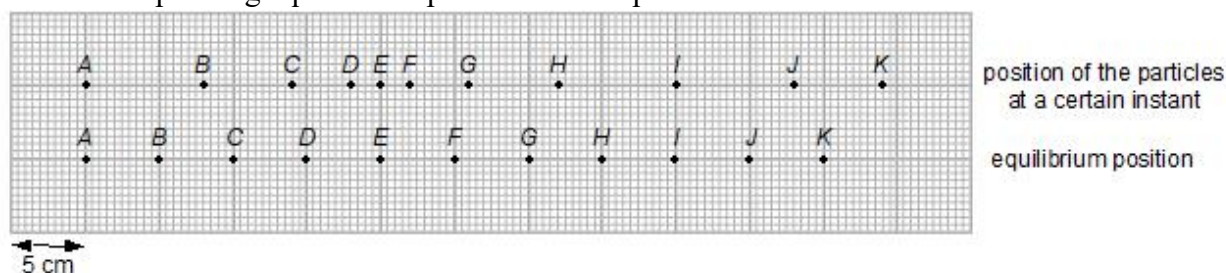
- (a) (i) Find the wavelength of the wave
(ii) Find the frequency of the wave
(iii) Find the distance between the vibrator and S.
- (b) State the bead(s) that is/are moving
- (i) In the same direction with T at time $t = 7$ s
(ii) In the opposite direction with T at time $t = 7$ s
- (c) In figure 12, sketch the waveform between P and T at a quarter of period after $t = 7$ s. Mark the position of S.

- (a) (i) $\lambda = 9 \text{ cm} / 2.25$
 $= 4 \text{ cm}$
- (ii) $T = (7 - 3) / 5 = 0.8 \text{ s}$
 $f = 1 / T$
 $= 1.25 \text{ Hz}$
- (iii) $v = f\lambda$
 $= 1.25 \times 4 = 5 \text{ cm s}^{-1}$
 $s = vt$
 $= 5 \times 3$
 $= 15 \text{ cm}$

- (b) (i) Q
(ii) S

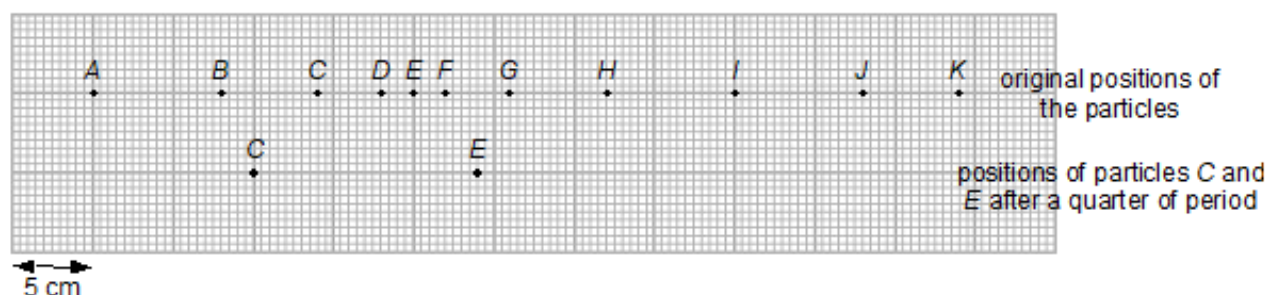


29. The figures below show the positions of some particles in a medium when a wave is passing through them, and the corresponding equilibrium positions of the particles.



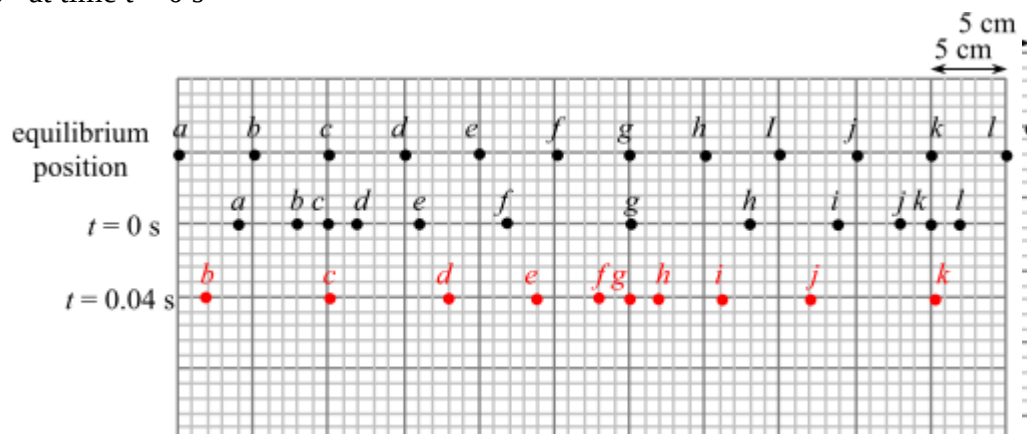
- State the type of wave travelling in the medium.
- Which particle(s) is/are
 - at the centre of a compression?
 - at the centre of a rarefaction?
 - at rest momentarily?
- What is the wavelength of the wave?
- It is known that each of the particles takes 0.2 s to complete one oscillation. Find the
 - frequency,
 - speed,
 - amplitude of the wave.
- Suppose the wave is travelling to the right. Draw the positions of particles C and E after a quarter of a period.

- Longitudinal wave (1A)⁺
- Particle E (1A)⁺
 - Particles A and I (1A)⁺
 - Particles C and K (1A)⁺
- The distance between particles E and I is equal to half a wavelength (1M).
Therefore, the wavelength is $\lambda = 2 \times 20 = 40$ cm (1A).⁺
- Frequency of the wave = $1/0.2 = 5$ Hz (1M+1A)⁺
 - Speed of the wave = $f\lambda = 5 \times 40 = 200$ m s⁻¹ (1M+1A)⁺
 - The amplitude of the wave is 4 cm. (1A)⁺
- ⁺

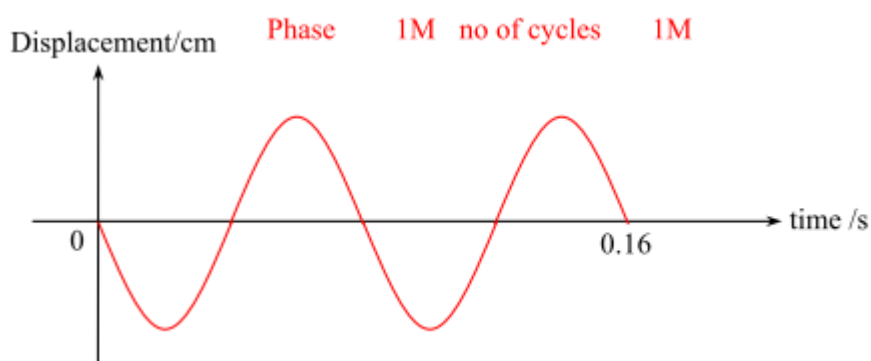


(1A).⁺

30. The figure below shows the positions of the particles in a longitudinal wave travelling to the right at a speed of 5 ms^{-1} at time $t = 0 \text{ s}$



- Write down the (i) amplitude and (ii) wavelength of the wave
- Describe the motion of the particles c , e and f at $t = 0 \text{ s}$
- Find the frequency of the wave
- Find the shortest time taken by particle e to reach its equilibrium position again from $t = 0 \text{ s}$
- Predict the wave pattern at $t = 0.04 \text{ s}$ and show your answers on the above figure
- Sketch the displacement-time graph of particle g from $t = 0 \text{ s}$ to $t = 0.16 \text{ s}$. (Take the direction to the right as positive)



- 4 cm (1 M)
 - 40 cm (1 M)
- c : Right 1M e : At rest 1M f : left 1M
- $f = 5/0.4$ 1M
 $= 12.5 \text{ Hz}$ 1M
- $T = 1/12.5 = 0.08 \text{ s}$ 1M
Time taken = $T/4 = 0.02 \text{ s}$ 1M